Full Stack Assignment – 05

Assignment ID: FSD-05

Title: Access Request Management Portal

Duration: 3 Hours

1️⃣ Objective

Design and develop an **Access Request Management Portal** that allows users to request access and approvers to review and act on those requests.

The system must enforce request constraints and ensure **data persistence using a database**.

2️⃣ User Roles

REQUESTER

- Register and log in to the system
- Submit an access request
- View own request status

APPROVER

- Log in to the system
- View all access requests
- Approve or reject access requests

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** at API and UI levels

Access Request Handling

- REQUESTER can submit access requests with required details
- All requests must be stored in the database
- APPROVER can approve or reject requests
- Request status must be retrieved from the database

4️⃣ Business Rule (Critical)

⚠️ **A user can have only one pending access request at any given time.**

If a pending request already exists, new requests must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All access request data **must persist in a database**
- Hardcoded or in-memory request storage is **not allowed**
- APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for REQUESTER and APPROVER
- Access request submission form
- Request listing with current status
- Approve/Reject actions for APPROVER

UI appearance is secondary; correctness and flow are evaluated.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions
- API endpoints

- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8️⃣ Evaluation Note

The assessment emphasizes **constraint handling, role enforcement, and end-to-end clarity**.